Lecture 8: Introduction to Linear Regression

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Suggested Reading: Hogg et al. 2018, Chapter 6.5 & 7.6.

So far we have discussed inference & testing on i.i.d. data. Starting from today, we move to the discussion on independent, but not necessarily identically distributed data. One common model for this data is the 'mean-shift' form, e.g., the observations $y_i = N(\mu_i, \sigma^2)$. However, without introducing models on μ_i, there are too many unknown parameters. A good way to simplify this is to give a 'model' to the mean function, and this becomes our regression problem.

1 Regression Problem: setting

A general regression problem supposes the observations are of the form

$$z_i = (x_{i1}, x_{i2}, \cdots, x_{ip}, y_i) \in \mathbf{R}^{p+1},\ i = 1, 2, \cdots, n,$$

$$i.e., \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} & y_1 \\ x_{21} & x_{22} & \cdots & x_{2p} & y_2 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} & y_n \end{bmatrix}$$

The purpose is to model mean of y_i based on x_i, i.e., we assume

$$\mu_i = \mathbf{E}y_i = f(x_{i1}, x_{i2}, \cdots, x_{ip})$$

and find the function f based on data.

Remark However, as you may ask, why this assumption is true, i.e., why y_i has to be a function of x_{ij}. Actually regression cannot answer this, it only finds the correlation between x and y. If you want to further find/explain the 'causal' relation between y and x_i, then you may have to learn the causal inference class, which gives you the answer.

We always call $(x_{i1}, x_{i2}, \cdots, x_{ip})$ the **explanatory** variables/independent variables, call y_i the **re-**

sponse/dependent variables. We may further write the matrix and the vector

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \cdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} \quad \text{and } \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \tag{1}$$

and call $\mathbf{X}$ the design matrix.

2 Regression problem: modelling

As statisticians, after collecting the data, we may want to estimate/inference/test f. To achieve this goal, we need to model the data. We adopt the fixed design setting here:

Definition 2.1 (Fixed design)**.** *Suppose the explanatory variables $x_{i1}, x_{i2}, \cdots, x_{ip}$ are fixed numbers, and the y_i are random variables satisfying $\mathbf{E}y_i = f(x_{i1}, \cdots, x_{ip})$.*

Remark Regression originated from analyzing experiments, in which the conditions $x_{i1}, \cdots, x_{ip}$ are fixed and designed (e.g., concentration $5\%, 10\%, 15\%$), that's why we consider the explanatory variables $x_{i1}, x_{i2}, \cdots, x_{ip}$ the fixed numbers. The favourable setting in the machine learning literature is the so-called random design, i.e., $x_{i1}, \cdots, x_{ip}$ are also random variables.

Estimating f without any assumptions are hard, so we may assign f to a parametric group before estimating it. A simple group is the linear function

$$\mathcal{F} = \left\{ f : \mathbf{R}^p \to \mathbf{R} : f(x_1, \cdots, x_p) = \sum_{j=1}^{p} x_j \beta_j, \beta_j \in \mathbf{R} \right\}$$

which is the main model considered in this class. Other example groups include the quadratic functions

$$\mathcal{G} = \left\{ g \in \mathbf{R}^p \to \mathbf{R} : g(x_1, \cdots, x_p) = \sum_{i=1}^{p} x_i \beta_i + \sum_{i=1}^{p} \sum_{j=1}^{p} \alpha_{ij} x_i x_j \right\}$$

and the trigonometric functions

$$\mathcal{H} = \left\{ h \in \mathbf{R}^p \to \mathbf{R} : h(x_1, \cdots, x_p) = \sum_{i=1}^{p} a_i \sin(x_i) + \sum_{j=1}^{p} b_j \cos(x_j) \right\}$$

3 Fitting a linear model: least-square (LS) method

Fitting the linear model actually does not involve too many statistical ideas. The only problem here is to find a $f \in \mathcal{F}$ that fits y_i well. To achieve this goal, we first need to define what is a 'good' model.

Definition 3.1. *We suppose a good linear model f should minimize the 'empirical square error'*

$$L(f;\{x_{i1},x_{i2},\cdots,x_{ip},y_i\}_{i=1}^n)=\frac{1}{n}\sum_{i=1}^n(y_i-f(x_{i1},x_{i2},\cdots,x_{ip}))^2,\ \ f\in\mathcal{F}.$$

The model f that minimizes the square error is called the **least-square** *fit.*

Since we assume $\mathcal{F}$ the group of linear functions, so

$$\min_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n(y_i-f(x_{i1},x_{i2},\cdots,x_{ip}))^2=\min_{\beta_1,\cdots,\beta_p\in\mathbf{R}}\frac{1}{n}\sum_{i=1}^n(y_i-\sum_{j=1}^p x_{ij}\beta_j)^2:=\min_{\beta_1,\cdots,\beta_p\in\mathbf{R}}L(\beta_1,\cdots,\beta_p)$$

By taking derivative, we have

$$\frac{\partial L}{\partial\beta_k}=0\Rightarrow\frac{1}{n}\sum_{i=1}^n x_{ik}y_i=\sum_{j=1}^p\left(\frac{1}{n}\sum_{i=1}^n x_{ij}x_{ik}\right)\times\beta_j\ \text{ for }\ \ k=1,2,\cdots,p.$$

Use the matrix representation in eq.(1), we have

$$\frac{1}{n}\mathbf{X}^T\mathbf{y}=(\frac{1}{n}\mathbf{X}^T\mathbf{X})\boldsymbol{\beta},\ \text{here}\ \ \boldsymbol{\beta}=(\beta_1,\beta_2,\cdots,\beta_p)^T\in\mathbf{R}^{p\times1}.$$

So, if this linear equality has solution, i.e., if the matrix $\mathbf{X}^T\mathbf{X}$ has rank p (invertible), then we have

$$\widehat{\boldsymbol{\beta}}=\underset{\beta}{\operatorname{argmin}}\,L(\boldsymbol{\beta})=(\frac{1}{n}\mathbf{X}^T\mathbf{X})^{-1}\left(\frac{1}{n}\mathbf{X}^T\mathbf{y}\right)=(\mathbf{X}^T\mathbf{X})^{-1}\left(\mathbf{X}^T\mathbf{y}\right).$$

Remark

i. If you want to include the constant term, i.e., set $f(x_1,\cdots,x_p)=\beta_0+\sum_{j=1}^p x_j\beta_j$, then you only need to consider the new design matrix

$$\mathbf{X}_1\in\mathbf{R}^{n\times(p+1)}=\begin{bmatrix}1 & x_{11} & x_{12} & \cdots & x_{1p}\\ 1 & x_{21} & x_{22} & \cdots & x_{2p}\\ \vdots & \vdots & \vdots & \cdots & \vdots\\ 1 & x_{n1} & x_{n2} & \cdots & x_{np}\end{bmatrix}$$

and derive the $p+1$ dimensional vector $\widehat{\boldsymbol{\beta}}_1=(\widehat{\beta}_0,\widehat{\beta}_1,\cdots,\widehat{\beta}_p)^T$ such that

$$\widehat{\boldsymbol{\beta}}_1=(\mathbf{X}_1^T\mathbf{X}_1)^{-1}\left(\mathbf{X}_1^T\mathbf{y}\right)$$

then the first term in $\widehat{\boldsymbol{\beta}}_1$ is the estimated coefficient for β_0.

ii. The 'empirical square error' is not a necessary error, e.g., you can consider minimizing the 'absolute

error'

$$L' = \frac{1}{n}\sum_{i=1}^{n} |y_i - f(x_{i1}, x_{i2}, \cdots, x_{ip})|$$

which is good if your data has large outlier.

4 Example: Fitting a linear model with 1 regressor

Here we suppose the observations are $(x_i, y_i), i = 1, \cdots, n$, $x_i, y_i \in \mathbf{R}$, and we fit the data with a linear model

$$\mathbf{E}y_i = f(x_i) = \beta_0 + x_i\beta_1,$$

β_0 is also called the intercept, and β_1 called the slope.

Direct calculation: The square error is

$$L(\beta_0, \beta_1) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \beta_0 - x_i\beta_1)^2.$$

Notations We first define some notations to simplify the derivations:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i,$$

$$T_{xx} = \sum_{i=1}^{n} x_i^2, \quad T_{xy} = \sum_{i=1}^{n} x_i y_i,$$

$$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 = T_{xx} - n\bar{x}^2, \quad S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = T_{xy} - n\bar{x}\bar{y}.$$

To find β_0, β_1 that minimizes the error function $L(\beta_0, \beta_1)$, we take derivative:

$$\frac{\partial L}{\partial \beta_0} = 0 \Rightarrow -\frac{1}{n}\sum_{i=1}^{n} 2(y_i - \beta_0 - x_i\beta_1) = 0 \Rightarrow \bar{y} - \beta_0 - \beta_1\bar{x} = 0$$
$$\frac{\partial L}{\partial \beta_1} = 0 \Rightarrow -\frac{1}{n}\sum_{i=1}^{n} 2x_i(y_i - \beta_0 - x_i\beta_1) = 0 \Rightarrow T_{xy} - n\beta_0\bar{x} - T_{xx}\beta_1 = 0$$

Solving the above two equations yields:

$$\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1\bar{x},$$
$$\widehat{\beta}_1 = \frac{T_{xy} - n\bar{x}\bar{y}}{T_{xx} - n\bar{x}^2} = \frac{S_{xy}}{S_{xx}}.$$

The alternative matrix form: In this situation, we have from Section 3 above:

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \Rightarrow \mathbf{X}^T\mathbf{X} = \begin{bmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{bmatrix} \text{ and } \mathbf{X}^T\mathbf{y} = \begin{bmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i y_i \end{bmatrix}$$

Then

$$\left(\mathbf{X}^T\mathbf{X}\right)^{-1} = \begin{bmatrix} n & n\bar{x} \\ n\bar{x} & T_{xx} \end{bmatrix}^{-1} = \frac{1}{nT_{xx} - n^2\bar{x}^2} \begin{bmatrix} T_{xx} & -n\bar{x} \\ -n\bar{x} & n \end{bmatrix} = \frac{1}{nS_{xx}} \begin{bmatrix} T_{xx} & -n\bar{x} \\ -n\bar{x} & n \end{bmatrix}$$

and thus

$$\begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \frac{1}{S_{xx}} \begin{bmatrix} T_{xx} & -n\bar{x} \\ -n\bar{x} & n \end{bmatrix} \begin{bmatrix} \bar{y} \\ \frac{1}{n}T_{xy} \end{bmatrix} = \frac{1}{S_{xx}} \begin{bmatrix} T_{xx}\bar{y} - T_{xy}\bar{x} \\ T_{xy} - n\bar{x}\bar{y} \end{bmatrix}$$

Which means that the

$$\widehat{\beta}_1 = \frac{T_{xy} - n\bar{x}\bar{y}}{S_{xx}} = \frac{S_{xy}}{S_{xx}}, \quad \widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1\bar{x}.$$

However, as you can see, it is very complex for $p > 1$ without matrix representation, so we recommend using matrix representation when you try to analyze the linear model.

5 Residuals in regression, and how to verify a linear model works

Recall our definition $\mathbf{E}y_i = f(x_{i1}, \cdots, x_{ip}) = \sum_{j=1}^p x_{ij}\beta_j$, so if we define the residual $\epsilon_i = y_i - f(x_{i1}, \cdots, x_{ip})$, then $\mathbf{E}\epsilon_i = 0$, and thus we usually write:

$$y_i = \sum_{j=1}^p x_{ij}\beta_j + \epsilon_i, \text{ or } \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \ \boldsymbol{\epsilon} = (\epsilon_1, \cdots, \epsilon_n)^T$$

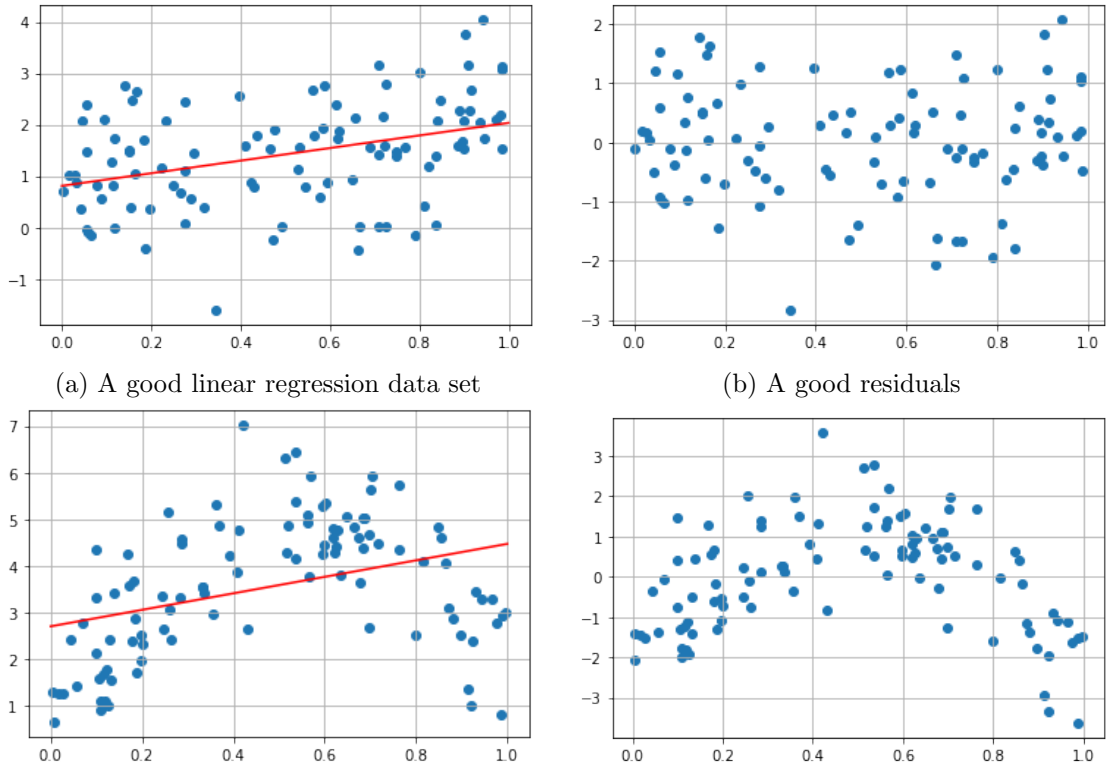
Remark For $\mathbf{E}\epsilon_i = 0$, we consider ϵ_i as non-informative, i.e., ϵ_i does not contain any useful information about the 'average effect' of y_i. This decomposition also describes our hope, i.e., the total effect y_i can be very complex, like figure 1 (c), but the effect can be separated into a simple 'mean effect', and a complex 'individual effect'. Moreover, the individual effect should cancels each other (zero-mean).

This hope actually gives us a hint on when to use a linear model, i.e., use the linear model if the

residuals

$$\hat{\epsilon}_i := y_i - \sum_{j=1}^{p} x_{ij}\widehat{\beta}_j, \quad i = 1, 2, \ldots, n$$

equally distributed around 0. To verify this, we can use the so-called 'regressor vs residuals' plot, whose x-axis are regressor, y-axis are residuals, to verify the model. If there is no clear trend in the regressor vs residuals plot, then we can consider it reasonable to use linear model to fit the data. Another idea is to use the cross-validation to validate the model. However, we will not introduce this in detail.



(a) A good linear regression data set

(b) A good residuals

(c) A bad linear regression data set

(d) A bad residuals

Figure 1: Examples of good and bad residuals.

For we do not want to further explain the complex individual effect, we prefer using a random variable to model the individual effect. To make it simple, we assume that the errors ϵ_i are i.i.d. $N(0, \sigma^2)$, in other words, y_i satisfies normal distribution $N(\sum_{j=1}^{p} x_{ij}\beta_j, \sigma^2)$, or write in matrix form: $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 I_n)$.

This definition has many advantages. For example if we use least-square estimator

$$\widehat{\boldsymbol{\beta}} = (\frac{1}{n}\mathbf{X}^T\mathbf{X})^{-1}\left(\frac{1}{n}\mathbf{X}^T\mathbf{y}\right)$$

then we can find the joint distribution of $\widehat{\boldsymbol{\beta}}$.

Fact: Suppose data $\boldsymbol{\epsilon} = (\epsilon_1, \epsilon_2, \cdots, \epsilon_n)^T$ have joint normal distribution, then the linear combinations $\mathbf{A}\boldsymbol{\epsilon}$ has joint normal distribution with mean $\mathbf{E}[\mathbf{A}\boldsymbol{\epsilon}] = \mathbf{A}\mathbf{E}[\boldsymbol{\epsilon}]$ and covariance $\mathrm{Cov}[\mathbf{A}\boldsymbol{\epsilon}] = \mathbf{A}\mathrm{Cov}(\boldsymbol{\epsilon})\mathbf{A}^T$.

For $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, we have the least-square estimator:

$$\widehat{\boldsymbol{\beta}} = (\frac{1}{n}\mathbf{X}^T\mathbf{X})^{-1}\left(\frac{1}{n}\mathbf{X}^T\mathbf{y}\right) = (\frac{1}{n}\mathbf{X}^T\mathbf{X})^{-1}\left(\frac{1}{n}\mathbf{X}^T\mathbf{X}\right)\boldsymbol{\beta} + (\frac{1}{n}\mathbf{X}^T\mathbf{X})^{-1}\frac{1}{n}\mathbf{X}^T\boldsymbol{\epsilon}$$
$$\Rightarrow \sqrt{n}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = (\frac{1}{n}\mathbf{X}^T\mathbf{X})^{-1}\frac{1}{\sqrt{n}}\mathbf{X}^T\boldsymbol{\epsilon}$$

which has normal distribution since $\boldsymbol{\epsilon}$ is assumed to be normal.

Define the matrix $\mathbf{H} = \{h_{ij}\}_{i=1,\cdots,p,j=1,\cdots,n} = (\frac{1}{n}\mathbf{X}^T\mathbf{X})^{-1}\frac{1}{\sqrt{n}}\mathbf{X}^T$, then

$$\sqrt{n}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \mathbf{H}\boldsymbol{\epsilon},$$

and its i-th entry is $\{\sum_{j=1}^n h_{ij}\epsilon_j\}$, which has joint normal distribution, and the mean is

$$\mathbf{E}\sum_{j=1}^n h_{ij}\epsilon_j = \sum_{j=1}^n h_{ij}\mathbf{E}\epsilon_j = 0$$

while the variance is

$$\mathrm{Var}(\sum_{j=1}^n h_{ij}\epsilon_j) = \mathbf{E}(\sum_{j=1}^n h_{ij}\epsilon_j)^2 = \sigma^2 \times \sum_{j=1}^n h_{ij}^2$$

This means

$$\sqrt{n}(\widehat{\beta}_i - \beta_i) \sim N(0, \sigma^2\sum_{j=1}^n h_{ij}^2)$$

Example with 1 regressor. For example, in the $p = 1$ example in Section 4, we have

$$\mathbf{H} = (\frac{1}{n}\mathbf{X}^T\mathbf{X})^{-1}\frac{1}{\sqrt{n}}\mathbf{X}^T = \frac{1}{S_{xx}}\begin{bmatrix} T_{xx} & -n\bar{x} \\ -n\bar{x} & n \end{bmatrix}\frac{1}{\sqrt{n}}\begin{bmatrix} 1 & \cdots & 1 \\ x_1 & \cdots & x_n \end{bmatrix}$$
$$= \frac{1}{\sqrt{n}S_{xx}}\begin{bmatrix} T_{xx} - n\bar{x}x_1 & \cdots & T_{xx} - n\bar{x}x_n \\ n(x_1 - \bar{x}) & \cdots & n(x_n - \bar{x}) \end{bmatrix}$$

Thus the variance of $\sqrt{n}(\widehat{\beta}_1 - \beta_1)$ is

$$\mathrm{Var}(\sqrt{n}(\widehat{\beta}_1 - \beta_1)) = \sigma^2\frac{1}{nS_{xx}^2}\sum_j n^2(x_j - \bar{x})^2 = \sigma^2\frac{n^2S_{xx}}{nS_{xx}^2} = \sigma^2\frac{n}{S_{xx}}.$$

and the variance of $\sqrt{n}(\widehat{\beta}_0 - \beta_0)$ is

$$\begin{aligned}
\mathrm{Var}(\sqrt{n}(\widehat{\beta}_0 - \beta_0)) &= \sigma^2 \frac{1}{nS_{xx}^2} \sum_j (T_{xx} - n\bar{x}x_j)^2 = \sigma^2 \frac{1}{nS_{xx}^2} \sum_j (S_{xx} + n\bar{x}\bar{x} - n\bar{x}x_j)^2 \\
&= \sigma^2 \frac{1}{nS_{xx}^2} \sum_j (S_{xx} + n\bar{x}(\bar{x} - x_j))^2 \\
&= \sigma^2 \frac{1}{nS_{xx}^2} \left[nS_{xx}^2 + n^2\bar{x}^2 \sum_j (\bar{x} - x_j)^2 \right] \\
&= \sigma^2 \frac{1}{nS_{xx}^2} \left[nS_{xx}^2 + n^2\bar{x}^2 S_{xx} \right] \\
&= \sigma^2 \left[1 + \frac{n\bar{x}^2}{S_{xx}} \right].
\end{aligned}$$

6 Maximum likelihood estimation

Suppose

$$y_i = \sum_{j=1}^{p} x_{ij}\beta_j + \epsilon_i, \text{ or } \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \ \boldsymbol{\epsilon} = (\epsilon_1, \cdots, \epsilon_n)^T,$$

where $\epsilon_i \sim N(0, \sigma^2)$, $i = 1, 2, \ldots, n$. In other words, y_i satisfies normal distribution $N(\sum_{j=1}^{p} x_{ij}\beta_j, \sigma^2)$.

We can also apply the maximum likelihood method to fit $\boldsymbol{\beta}$. To do that, note that

$$\begin{aligned}
y_i = \sum_{j=1}^{p} x_{ij}\beta_j + \epsilon_i \sim N(\sum_{j=1}^{p} x_{ij}\beta_j, \sigma^2) \\
\Rightarrow \text{Likelihood } f(y_1, \cdots y_n) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \sum_{j=1}^{p} x_{ij}\beta_j)^2 \right) \\
\Rightarrow \text{ Log-likelihood } \ell(\beta_1, \cdots, \beta_p, \sigma^2) = -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \sum_{j=1}^{p} x_{ij}\beta_j)^2 - \frac{n}{2} \log(2\pi\sigma^2)
\end{aligned}$$

Taking the gradient, we have

$$\begin{aligned}
\frac{\partial L}{\partial \beta_k} = 0 \Rightarrow \sum_{i=1}^{n} (y_i - \sum_{j=1}^{p} x_{ij}\beta_j) \times (-x_{ik}) = 0 \\
\frac{\partial L}{\partial (\sigma^2)} = 0 \Rightarrow \sigma^2 = \frac{1}{n} \sum_{i=1}^{n} (y_i - \sum_{j=1}^{p} x_{ij}\beta_j)^2
\end{aligned}$$

Therefore, the maximum-likelihood estimator of β, σ^2 is

$$\widehat{\beta} = (\mathbf{X^T X})^{-1} \mathbf{X^T y}, \text{ and } \widehat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (y_i - \sum_{j=1}^{p} x_{ij}\widehat{\beta}_j)^2$$

which coincides with the previous deduction.

7 A proof of Consistency of σ^2 (error variance)*

In the previous variance calculations in Section 5, you can see that finding σ^2 is a crucial thing since it directly affects the variance of the LS estimate $\widehat{\beta}$ and $\widehat{\beta}_1$, thus the width of confidence interval (introduced later). When σ^2 is unknown, we can use the MLE estimate $\sigma^2 = \frac{1}{n}\sum_{i=1}^n (y_i - \sum_{j=1}^n x_{ij}\widehat{\beta}_j)^2$ in Section 6. Recall that from Lecture 2, MLE are consistent (we have the MLE $\hat{\theta}$ approximately follow a normal distribution with mean θ).

This section gives a detailed proof for the consistency of σ^2 under the example of linear regression. This part is optional and is NOT required.

Define the estimator of residuals

$$\widehat{\epsilon}_i = y_i - \sum_{j=1}^p x_{ij}\widehat{\beta}_j \quad \text{and} \quad \widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n \widehat{\epsilon}_i^2,$$

we then prove the consistency of the variance estimator $\widehat{\sigma}^2$.

Theorem 7.1. *Suppose observations* $(x_{i1}, x_{i2}, \cdots, x_{ip}, y_i), i = 1, \cdots, n$ *satisfy*

$$y_i = \sum_{j=1}^p x_{ij}\beta_j + \epsilon_i, \text{ here } \epsilon_i \sim N(0, \sigma^2)$$

then the estimator

$$\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n \widehat{\epsilon}_i^2, \quad \text{here } \widehat{\epsilon}_i = y_i - \sum_{j=1}^p x_{ij}\widehat{\beta}_j \tag{2}$$

satisfies

$$\widehat{\sigma}^2 \to_p \sigma^2 \text{ as } n \to \infty.$$

Proof Define the vector $\mathbf{x}_i = (x_{i1}, x_{i2}, \cdots, x_{ip})^T$, $\boldsymbol{\beta} = (\beta_1, \cdots, \beta_p)^T$, and $\widehat{\boldsymbol{\beta}} = (\widehat{\beta}_1, \cdots, \widehat{\beta}_p)^T$, then we have

$$\widehat{\epsilon}_i = y_i - \mathbf{x}_i^T\widehat{\boldsymbol{\beta}} = \epsilon_i - \mathbf{x}_i^T(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})$$

so that

$$\frac{1}{n}\sum_{i=1}^n \widehat{\epsilon}_i^2 = \frac{1}{n}\sum_{i=1}^n \epsilon_i^2 + \frac{1}{n}\sum_{i=1}^n (\mathbf{x}_i^T(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}))^2 - \frac{2}{n}\sum_{i=1}^n \epsilon_i \mathbf{x}_i^T(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})$$

From the law of large number,

$$\frac{1}{n}\sum_{i=1}^n \epsilon_i^2 \to_p \mathbf{E}\epsilon_1^2 = \sigma^2$$

Therefore, we only need to show the second and third terms both converge to 0.

For the second term,

$$\frac{1}{n}\sum_{i=1}^{n}(\mathbf{x}_i^T(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta}))^2=(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta})^T\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{x}_i^T(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta})=(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta})^T\frac{\mathbf{X}^T\mathbf{X}}{n}(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta})$$

Since $\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta}=(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon}$,

$$(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta})^T\frac{\mathbf{X}^T\mathbf{X}}{n}(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta})=\frac{1}{n}\boldsymbol{\epsilon}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon}$$

Notice that $\boldsymbol{\epsilon}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon}$ is a scalar, thus

$$\mathbf{E}\boldsymbol{\epsilon}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon}=\mathbf{E}[Tr(\boldsymbol{\epsilon}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon})].$$

Since $\mathbf{E}[Tr(AB)]=\mathbf{E}[Tr(BA)]$ for any matrix A,B, we have

$$\begin{aligned}\mathbf{E}\boldsymbol{\epsilon}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon}&=\mathbf{E}[Tr(\boldsymbol{\epsilon}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon})]\\&=\mathbf{E}[Tr(\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T)]\\&=Tr(\mathbf{E}[\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T])\\&=Tr(\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T])\\&=\sigma^2\times Tr(\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T)=\sigma^2\times Tr((\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X})=\sigma^2 p,\end{aligned}$$

where we have used the fact that $\mathbf{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T]=\sigma^2 I_n$.

From Markov inequality $(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta})^T\frac{\mathbf{X}^T\mathbf{X}}{n}(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta})\to_p 0$.

Finally, from Cauchy–Schwarz inequality,

$$\begin{aligned}|\frac{1}{n}\sum_{i=1}^{n}\epsilon_i\mathbf{x}_i^T(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta})|^2&\le\frac{1}{n}\sum_{i=1}^{n}\epsilon_i^2\times\frac{1}{n}\sum_{i=1}^{n}|\mathbf{x}_i^T(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta})|^2\\\text{and }\frac{1}{n}\sum_{i=1}^{n}\epsilon_i^2\times\frac{1}{n}\sum_{i=1}^{n}|\mathbf{x}_i^T(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta})|^2&\to_p\sigma^2\times 0=0\\\text{so }|\frac{1}{n}\sum_{i=1}^{n}\epsilon_i\mathbf{x}_i^T(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta})|^2&\to_p 0\end{aligned}$$

and we prove the result. □

After estimating the error's variance σ^2, we are able to establish confidence interval for the coefficients using the following steps:

i. Fit the linear model, i.e., calculate the coefficients $\widehat{\beta}$. Then calculate the matrix $\mathbf{H}=\{h_{ij}\}_{i=1,\cdots,p,j=1,\cdots,n}=(\frac{1}{n}\mathbf{X}^T\mathbf{X})^{-1}\frac{1}{\sqrt{n}}\mathbf{X}^T$.

ii. Estimating the errors' variance $\widehat{\sigma}^2$ using eq.(2).

ii. Then the (two-sided) $1-\alpha$ confidence interval for β_i is

$$\left[\widehat{\beta}_i - \frac{\widehat{\sigma}}{\sqrt{n}}\sqrt{\sum_{j=1}^{n} h_{ij}^2} \times z_{\alpha/2},\ \widehat{\beta}_i + \frac{\widehat{\sigma}}{\sqrt{n}}\sqrt{\sum_{j=1}^{n} h_{ij}^2} \times z_{\alpha/2}\right]$$

From Slutsky theorem, we know that this interval has asymptotic coverage probability $1-\alpha$.

References

Hogg, Robert V., Elliot A. Tanis, and Dale L. Zimmerman (2018). *Probability and statistical inference.*